

FINAL EXPLANATION: MATHEMATICS GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: MATHEMATICS GRADE 10 | SUB-STRAND 2.1: SIMILARITY AND ENLARGEMENT | FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

Driving question: How can a full-size photograph of a person be resized to fit a small passport photo box without distortion? Remember Wanjiru Atieno of Nyalenda Secondary School, Kisumu, and her four prints from Mr Otieno at Lakeview Photo Studio: the original $14\text{ cm} \times 18\text{ cm}$, the passport photo $3.5\text{ cm} \times 4.5\text{ cm}$, the prize-giving poster $70\text{ cm} \times 90\text{ cm}$, and the $4.5\text{ cm} \times 4.5\text{ cm}$ trial print made by changing only the width. Answer each section in the space under it. Use your own measurements and calculations from Lessons 1 to 12. Write the claim first, then evidence (numbers), then reasoning (why the numbers matter). Use k for the linear scale factor, $k = \text{image length} \div \text{object length}$, and always say which print is the object and which is the image. Write width first, then height. Show your working and include units.

Section 1: Claim and evidence from the ratios and angles

Use the table below, which comes from your measurements of the four prints.

Print	Width $\times$ height (cm)	Width : height	Chin angle
Original	14×18	7 : 9	about 42.5°
Passport	3.5×4.5	7 : 9	about 42.5°
Poster	70×90	7 : 9	about 42.5°

(a) Claim: the passport photo ($3.5\text{ cm} \times 4.5\text{ cm}$) and the poster ($70\text{ cm} \times 90\text{ cm}$) keep Wanjiru's likeness because each is similar (~) to the original $14\text{ cm} \times 18\text{ cm}$ print. The $4.5\text{ cm} \times 4.5\text{ cm}$ trial print does not keep her likeness; her face looks stretched.

(b) Evidence: the original, passport and poster all have a width : height ratio of 7 : 9 ($14 : 18$, $3.5 : 4.5$ and $70 : 90$ all simplify to 7 : 9). Their chin angles are all about 42.5° , and all their corners are 90° . The trial print has width : height = 1 : 1, and its chin angle is about 53° , so both the ratio and the angle changed.

(c) Reasoning: figures are similar (~) when all corresponding angles are equal and the ratios of corresponding sides are equal. The passport photo and poster pass both tests, so only their size changed. The trial print fails both tests, so its shape changed. What must stay the same is the shape (equal corresponding angles and equal ratios of corresponding sides). What may change is the size.

			°
Trial print	4.5×4.5	1 : 1	about 53°

(a) Write a claim: which prints keep Wanjiru's likeness, and which print does not? (b) Use the width : height ratios and the chin angles to give evidence for your claim. (c) Explain, using the words similar (~), corresponding sides and corresponding angles, what must stay the same when a picture changes size.

Section 2: Scale factors

Take the original 14 cm × 18 cm print as the object. (a) Calculate the linear scale factor k for the passport photo (image) for width and for height. State whether this is an enlargement or a reduction. (b) Calculate k for the poster (image), for width and for height. (c) Find k from the passport photo (object) to the poster (image). (d) For the 4.5 cm × 4.5 cm trial print (image), calculate the width factor and the height factor from the original. Compare them and explain what your comparison tells you about the face in this print.

- (a) Object = original, image = passport. Width: $k = 3.5 \div 14 = 1/4$. Height: $k = 4.5 \div 18 = 1/4$. Both factors are equal to $1/4$ (0.25). Since $0 < k < 1$, this is a reduction.
- (b) Object = original, image = poster. Width: $k = 70 \div 14 = 5$. Height: $k = 90 \div 18 = 5$. Both are 5, and since $k > 1$ this is an enlargement.
- (c) Object = passport, image = poster: $k = 70 \div 3.5 = 20$, and $90 \div 4.5 = 20$. So $k = 20$.
- (d) Object = original, image = trial print. Width factor = $4.5 \div 14 = 9/28$ (about 0.32). Height factor = $4.5 \div 18 = 1/4$ (0.25). The two factors are not equal, so the width and height were changed by different amounts. The print is not similar to the original, so Wanjiru's face looks stretched. A single scale factor, applied to both width and height, is needed to keep the shape.

Section 3: Using scale factor and similar triangles to find unknown lengths

(a) In the original print the face triangle has an eye gap of 5.6 cm and a chin distance of 7.2 cm. Use k from the original to the passport photo to find the eye gap and the chin distance on the passport photo. **(b)** Now find the eye gap and the chin distance on the poster. **(c)** At Nyalenda Secondary School, at the same time of day, Wanjiru (1.5 m tall) casts a 2 m shadow and a tree casts a 12 m shadow. Explain why the two shadow triangles can be treated as similar, then find the height of the tree. **(d)** Explain how this problem is related to Wanjiru's photo.

- (a) Object = original, image = passport, $k = 1/4$. Eye gap = $1/4 \times 5.6 = 1.4$ cm. Chin distance = $1/4 \times 7.2 = 1.8$ cm.
 (b) Object = original, image = poster, $k = 5$. Eye gap = $5 \times 5.6 = 28$ cm. Chin distance = $5 \times 7.2 = 36$ cm.
 (c) The sun's rays make the same angle with the ground for Wanjiru and for the tree, and both stand at 90° to the ground, so the corresponding angles are equal and the triangles are similar ($\sim$). Take Wanjiru's triangle as the object and the tree's triangle as the image: $k = 12 \div 2 = 6$. Tree height = $1.5 \times 6 = 9$ m. (Check: $1.5 \div 2 = 0.75$, and $0.75 \times 12 = 9$ m.)
 (d) The same idea applies: when corresponding angles are equal, every length is multiplied by the same k . This keeps the face triangle in the passport photo ($k = 1/4$) and the poster ($k = 5$) the same shape as the original, and it lets us find a height we cannot measure directly.

Section 4: Area and volume factors

(a) Calculate the area of the original print, the passport photo and the poster. **(b)** Use your areas and the linear scale factors to find the area factor from the original to the passport photo and from the original to the poster. Write the pattern that connects the area factor to k . **(c)** Mr Otieno packs prints in a small carton of 10 cm $\times$ 6 cm $\times$ 4 cm and a larger similar carton of 20 cm $\times$ 12 cm $\times$ 8 cm. Find k , the volumes of both cartons and the volume factor. Write the pattern that connects the volume factor to k . **(d)** A friend says: 'If every length of the poster is multiplied by 5, then its area is multiplied by 5.' Explain whether your friend is correct.

- (a) Original: $14 \times 18 = 252$ cm². Passport: $3.5 \times 4.5 = 15.75$ cm². Poster: $70 \times 90 = 6300$ cm².
 (b) Original to passport: $k = 1/4$, so the area factor = $(1/4)^2 = 1/16$. Check: $15.75 \div 252 = 1/16$. Original to poster: $k = 5$, so the area factor = $5^2 = 25$. Check: $6300 \div 252 = 25$. Pattern: area factor = k^2 .
 (c) Small carton = object, large carton = image: $k = 20 \div 10 = 12 \div 6 = 8 \div 4 = 2$. Small carton volume = $10 \times 6 \times 4 = 240$ cm³. Large carton volume = $20 \times 12 \times 8 = 1920$ cm³. Volume factor = $1920 \div 240 = 8 = 2^3$. Pattern: volume factor = k^3 .
 (d) Your friend is not correct. Area depends on two lengths (width $\times$ height), and both are multiplied by 5, so the area is multiplied by $5 \times 5 = 5^2 = 25$. This is shown by $6300 \div 252 = 25$. The k^2 rule only applies when one scale factor is used for all lengths, that is, for similar figures.

Section 5: Enlargement, applications and the final explanation

(a) Triangle ABC has A(1,1), B(3,1) and C(1,2). Write the image coordinates for an enlargement with centre O(0,0) and scale factor 2. Explain how the rays from O are used to build the image. **(b)** Triangle PQR with P(2,2), Q(3,2), R(2,3) has the image P'(4,4), Q'(7,4), R'(4,7). Explain how to find the centre of the enlargement and find k. **(c)** On a 1:5 000 000 map, Kisumu and Nairobi are about 265 km apart in a straight line. Find their distance on the map in cm. **(d)** A school sign of 70 cm × 90 cm (0.63 m²) is enlarged with k = 3. Paint costs KES 400 per m². Find the cost of paint for each sign. **(e)** Now answer the driving question: How can a full-size photograph be resized to fit a small passport photo box without distortion? Also say why we need to enlarge or reduce objects, and how we use similarity and enlargement in day-to-day life.

(a) Each image point lies on the ray from O through the object point, at twice the distance from O. A'(2,2), B'(6,2), C'(2,4). The image is similar to the object: corresponding angles are equal and corresponding sides are in the ratio 1 : 2.

(b) Join each object point to its image (P to P', Q to Q', R to R') and extend the lines. They meet at one point, (1,1), which is the centre. $k = \text{image length} \div \text{object length} = 3 \div 1 = 3$ ($P'Q' = 3$, $PQ = 1$). Check: P' minus centre = (3,3) = $3 \times (1,1)$, which is $3 \times (P \text{ minus centre})$.

(c) $265 \text{ km} = 26\,500\,000 \text{ cm}$. Map distance = $26\,500\,000 \div 5\,000\,000 = 5.3 \text{ cm}$.

(d) Enlarged sign = $210 \text{ cm} \times 270 \text{ cm}$. Area factor = $3^2 = 9$, so area = $9 \times 0.63 = 5.67 \text{ m}^2$. Small sign paint: $0.63 \times 400 = \text{KES } 252$. Large sign paint: $5.67 \times 400 = \text{KES } 2268$, which is 9 times as much. The cost depends on area, so it changes by k^2 , not by k.

(e) Claim: a full-size photograph can be resized to fit the passport box without distortion by multiplying every length, both width and height, by the same scale factor. Evidence: the original $14 \text{ cm} \times 18 \text{ cm}$ becomes the passport photo $3.5 \text{ cm} \times 4.5 \text{ cm}$ with $k = 1/4$ for both width and height, and the poster $70 \text{ cm} \times 90 \text{ cm}$ with $k = 5$. All three have width : height = 7 : 9, 90° corners and a chin angle of about 42.5°. The $4.5 \text{ cm} \times 4.5 \text{ cm}$ print used a width factor of $9/28$ and a height factor of $1/4$, so its ratio became 1 : 1 and its chin angle about 53°. Reasoning: with one k the figures are similar (~), so corresponding angles and ratios of corresponding sides stay equal and only the size changes. Different factors change the shape and give distortion. Size can change by any scale factor, but shape is kept only if one scale factor is applied to all lengths in every direction. Area then changes by k^2 and volume by k^3 . We enlarge or reduce objects to fit them into a space or make them easier to see or handle: passport photos, posters, maps, models and painted signs. We use similarity and enlargement every day in maps, scale drawings, models, measuring heights from shadows and costing paint.

RUBRIC

Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Claim, evidence and reasoning	States a clear claim, supports it with exact data (7:9 vs 1:1, about 42.5° vs about 53°, $k = 1/4$ and 5 vs $9/28$ and $1/4$) and links the evidence to the idea of similarity.	States a correct claim and uses some correct data, but the reasoning linking the data to similarity is incomplete.	Claim is unclear or missing, or uses little or incorrect evidence.
Scale factor calculations	Calculates every k as image $\div$ object, names the object and the image, and correctly identifies enlargement or	Calculates most scale factors correctly but sometimes omits the object and image or the enlargement/reduction	Calculations are mostly incorrect or the object and image are mixed up.

	reduction. Compares the unequal factors of the trial print.	label.	
Area and volume factors	Correctly uses k^2 and k^3 with Wanjiru's prints and the cartons, and explains why doubling lengths does not double the area.	Uses k^2 and k^3 correctly in most cases but explanations are brief.	Treats area or volume as changing by k , or makes errors in the calculations.
Enlargement, applications and mathematical language	Finds images and the centre correctly, solves the map, sign cost and shadow problems accurately, and uses $\sim$, corresponding sides and corresponding angles, units and width-first order correctly.	Solves most application problems correctly with minor errors in units, notation or the order of width and height.	Has many errors in the applications, or little correct use of terms and units.
Answering the driving question	Gives a complete explanation that size can change but shape is kept only if one scale factor is applied to all lengths. Connects to the Key Inquiry Questions with real examples.	Gives a correct answer to the driving question but with limited links to everyday examples.	Answer is incomplete or does not explain why the trial print is distorted.

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